

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-KC-4501-07 Sheet 1 of 1
OPL Title	Intercooler EA-4502 Performance Check	Discipline	Mechanical	
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	Area / Unit	4500 - RECYCLE GAS SYSTEM	
Related Interlock	SEQ-4501 (SIL 2)	P&ID Ref	TJC-LLD-PID-4501	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **intercooler ea-4502 performance check** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Check the interstage temperature drop across EA-4502	Isolation & LOTO verified
2	Verify the jacket water flow FSL-4508 is adequate	Parts match BOM / spec
3	Clean the intercooler tubes if the approach temperature rises	Torque / clearance to spec
4	Log the interstage pressure and compare to datasheet	No leak / smooth operation
5	Report fouling trends so cleaning can be planned	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Poor intercooling raises the 2nd-stage suction temperature and hence the final discharge temperature towards the trip.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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